

Module - II

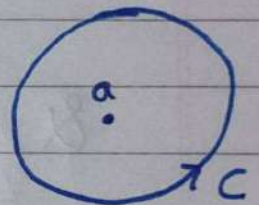
Complex Analysis II

Taylor's Series

If $f(z)$ is analytic inside a circle C with centre at a then for all z inside C ,

$$f(z) = f(a) + \frac{(z-a)}{1!} f'(a) + \frac{(z-a)^2}{2!} f''(a) + \dots + \frac{(z-a)^n}{n!} f^n(a)$$

OR



$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n, \text{ where } a_n = \frac{f^n(a)}{n!}.$$

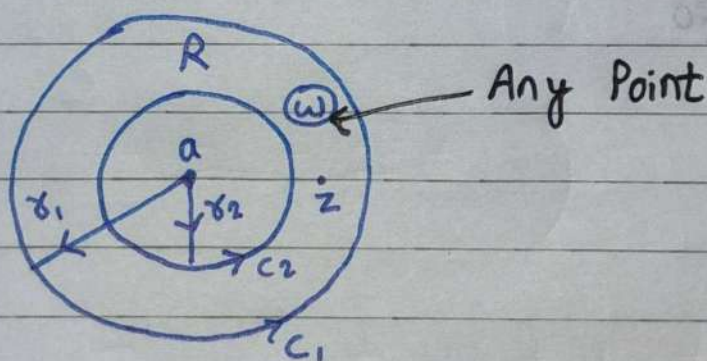
Laurent's Series

If $f(z)$ is analytic inside and on a boundary of an annular (ring shaped) region R bounded by two concentric circles C_1 & C_2 of radii r_1 & r_2 [$r_1 > r_2$] respectively having centre a , then for all z in R ,

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

$$\text{where, } a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)^{n+1}} dw \quad [n=0, 1, 2, 3, \dots]$$

$$\& \quad b_n = \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w-a)^{n+1}} dw \quad [n=1, 2, 3, \dots]$$



Expansion Formulae

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots, \quad |x| < 1$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots, \quad |x| < 1$$

Q.1) Expand $\frac{1}{z^2 - 3z + 2}$ in the region,

(i) $1 < |z| < 2$, (ii) $0 < |z-1| < 1$.

Solⁿ:

$$f(z) = \frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)} = \frac{A}{(z-1)} + \frac{B}{(z-2)}$$

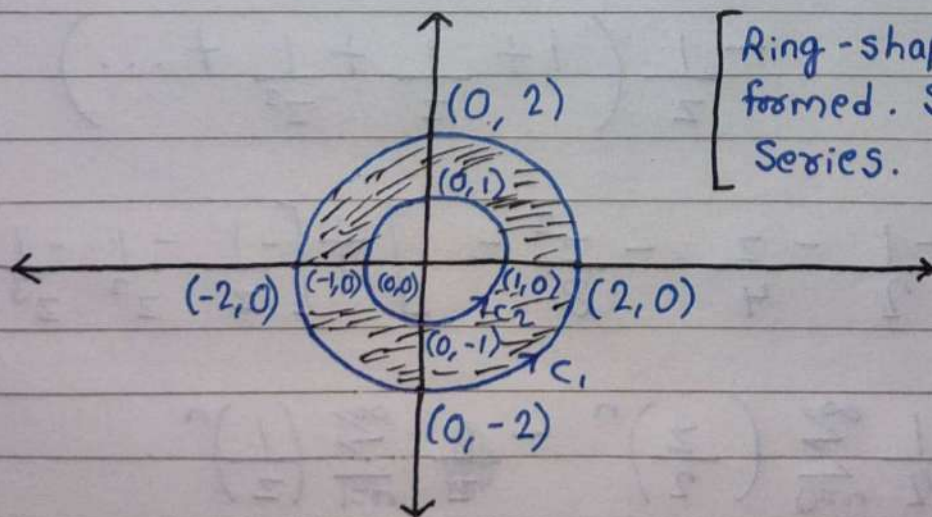
$$\Rightarrow 1 = A(z-2) + B(z-1)$$

At $z=2$,
 $B=1$

At $z=1$,
 $A=-1$

$$f(z) = \frac{1}{(z-2)} - \frac{1}{(z-1)}$$

(i) $1 < |z| < 2$



[Ring-shaped region is formed. So using Laurent's Series.]

We have,

$$f(z) = \frac{1}{(z-2)} - \frac{1}{(z-1)}$$

$$\Rightarrow f(z) = (z-2)^{-1} - (z-1)^{-1}$$

Given, $1 < |z| < 2$

i.e. $|z| > 1 \Rightarrow \left| \frac{1}{z} \right| < 1$

& $|z| < 2 \Rightarrow \left| \frac{z}{2} \right| < 1$

$\therefore \left| \frac{z}{2} \right| < 1$ & $\left| \frac{1}{z} \right| < 1$

$$\therefore f(z) = (-2)^{-1} \left[1 - \frac{z}{2} \right]^{-1} - z^{-1} \left[1 - \frac{1}{z} \right]^{-1}$$

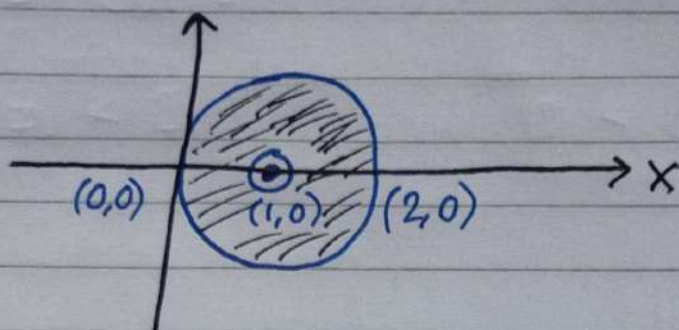
$$\Rightarrow f(z) = -\frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \dots \right)$$

$$- \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right)$$

$$\Rightarrow f(z) = \left[-\frac{1}{2} - \frac{z}{4} - \frac{z^2}{8} - \dots \right] + \left[-\frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3} - \dots \right]$$

$$\Rightarrow f(z) = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2} \right)^n - \sum_{n=1}^{\infty} \left(\frac{1}{z} \right)^n$$

(ii) $0 < |z-1| < 1$



We have,

$$f(z) = (z-2)^{-1} - (z-1)^{-1}$$

$$\Rightarrow f(z) = [(z-1)-1]^{-1} - (z-1)^{-1}$$

$$\because |z-1| < 1$$

$$\therefore f(z) = -[1 - (z-1)]^{-1} - (z-1)^{-1}$$

$$\Rightarrow f(z) = -[1 + (z-1) + (z-1)^2 + (z-1)^3 + \dots] - \frac{1}{(z-1)}$$

$$\Rightarrow f(z) = -\frac{1}{(z-1)} - 1 - (z-1) - (z-1)^2 - (z-1)^3 - \dots$$

$$\Rightarrow f(z) = \frac{-1}{(z-1)} - \sum_{n=0}^{\infty} (z-1)^n$$

Q.2) Expand, $f(z) = \frac{z}{(z-1)(2-z)}$ in Laurent series valid for (a) $|z-1| > 1$ & (b) $0 < |z-2| < 1$.

Solⁿ:

$$f(z) = \frac{z}{(z-1)(2-z)} = \frac{A}{(z-1)} + \frac{B}{(2-z)}$$

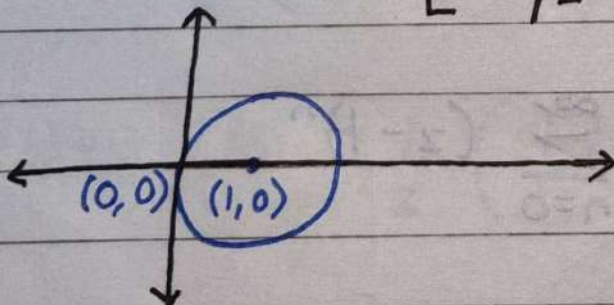
$$\Rightarrow z = A(2-z) + B(z-1)$$

$$\text{For } z=1, \\ A=1$$

$$\text{For } z=2, \\ B=2$$

$$f(z) = \frac{1}{(z-1)} + \frac{2}{(2-z)}$$

$$(a) \quad |z-1| > 1 \quad \left[\Rightarrow \left| \frac{1}{z-1} \right| < 1 \right]$$



$$f(z) = \frac{1}{(z-1)} + \frac{2}{[1-(z-1)]}$$

$$\Rightarrow f(z) = \frac{1}{(z-1)} + 2[1 - (z-1)]^{-1}$$

$$\therefore \left| \frac{1}{z-1} \right| < 1$$

$$\therefore f(z) = \frac{1}{(z-1)} - \frac{2}{(z-1)} \left[1 - \frac{1}{(z-1)} \right]^{-1}$$

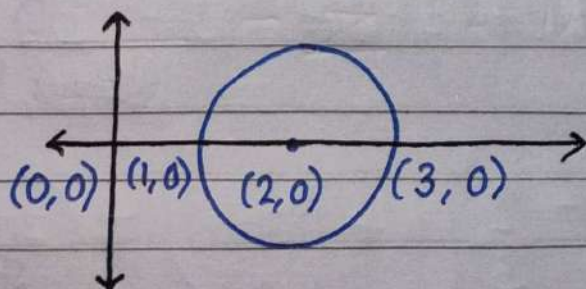
$$\Rightarrow f(z) = \frac{1}{(z-1)} - \frac{2}{(z-1)} \left[1 + \frac{1}{(z-1)} + \frac{1}{(z-1)^2} + \dots \right]$$

$$= \frac{1}{(z-1)} - \frac{2}{(z-1)} - \frac{2}{(z-1)^2} - \frac{2}{(z-1)^3} - \dots$$

$$= \frac{-1}{(z-1)} - \frac{2}{(z-1)^2} \left[1 + \frac{1}{(z-1)} + \frac{1}{(z-1)^2} + \dots \right]$$

$$f(z) = \frac{-1}{(z-1)} - \frac{2}{(z-1)} \sum_{n=0}^{\infty} \left[\frac{1}{(z-1)} \right]^n$$

(b) $0 < |z-2| < 1$



$$f(z) = \frac{1}{(z-1)} + \frac{2}{(2-z)}$$

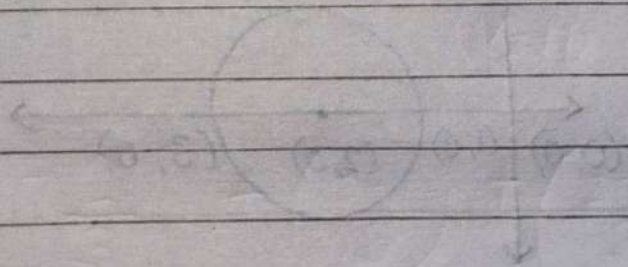
$$\Rightarrow f(z) = \frac{1}{[(z-2)+1]} - \frac{2}{(z-2)}$$

$$\Rightarrow f(z) = [1 + (z-2)]^{-1} - \frac{2}{(z-2)}$$

$$\therefore |z-2| < 1$$

$$\therefore f(z) = 1 - (z-2) + (z-2)^2 - (z-2)^3 - \dots - \frac{2}{(z-2)}$$

$$\Rightarrow f(z) = \sum_{n=0}^{\infty} (-1)^n (z-2)^n - \frac{2}{(z-2)}$$



Q.3) Find the Laurent's Expansion for $f(z) = \frac{7z-2}{z^3-z^2-2z}$

in the region given by:

(i) $0 < |z+1| < 1$

(ii) $1 < |z+1| < 3$

(iii) $|z+1| > 3$

Soln:

$$f(z) = \frac{7z-2}{z^3-z^2-2z} = \frac{7z-2}{z(z^2-z-2)} = \frac{7z-2}{z(z+1)(z-2)}$$

$$= \frac{A}{z} + \frac{B}{(z+1)} + \frac{C}{(z-2)}$$

$$\Rightarrow 7z-2 = A(z+1)(z-2) + B(z)(z-2) + C(z)(z+1)$$

At $z=0$,
 $-2 = -2A$

$$\Rightarrow A = -1$$

At $z=2$,

$$12 = 6C$$

$$\Rightarrow C = 2$$

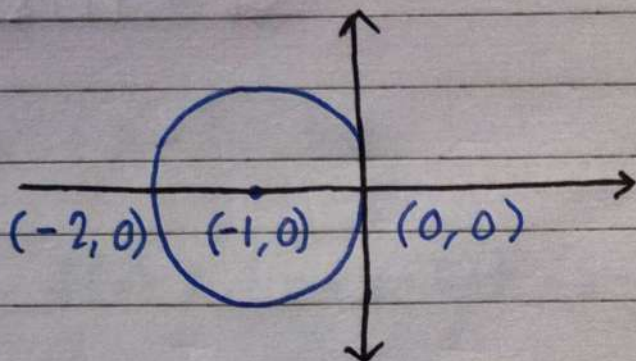
At $z=-1$,

$$-9 = 3B$$

$$\Rightarrow B = -3$$

$$f(z) = \frac{1}{z} - \frac{3}{(z+1)} + \frac{2}{(z-2)}$$

(i) $0 < |z+1| < 1$



$$f(z) = \frac{1}{[(z+1)-1]} - \frac{3}{(z+1)} + \frac{2}{[(z+1)-3]}$$

$$\Rightarrow f(z) = [(z+1)-1]^{-1} - \frac{3}{(z+1)} + 2[(z+1)-3]^{-1}$$

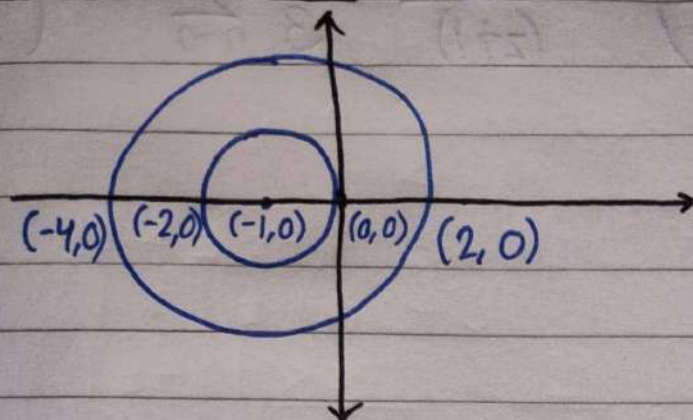
$$\therefore |z+1| < 1 \Rightarrow \left| \frac{z+1}{3} \right| < \frac{1}{3} < 1$$

$$\therefore f(z) = -[1 - (z+1)]^{-1} - \frac{3}{(z+1)} - \frac{2}{3} \left[1 - \left(\frac{z+1}{3} \right) \right]^{-1}$$

$$\Rightarrow f(z) = -[1 + (z+1) + (z+1)^2 + \dots] - \frac{3}{(z+1)}$$

$$- \frac{2}{3} \left[1 + \left(\frac{z+1}{3} \right) + \left(\frac{z+1}{3} \right)^2 + \dots \right]$$

(ii) $1 < |z+1| < 3$



$$f(z) = \frac{1}{z} - \frac{3}{(z+1)} + \frac{2}{(z-2)}$$

$$\Rightarrow f(z) = [(z+1)-1]^{-1} - \frac{3}{(z+1)} + 2[(z+1)-3]^{-1}$$

$$\because \left| \frac{1}{z+1} \right| < 1 \quad \& \quad \left| \frac{z+1}{3} \right| < 1$$

$$\therefore f(z) = \frac{-1}{(z+1)} \left[1 - \left(\frac{1}{z+1} \right) \right]^{-1} - \frac{3}{(z+1)} - \frac{2}{3} \left[1 - \left(\frac{z+1}{3} \right) \right]^{-1}$$

$$\Rightarrow f(z) = \frac{-1}{(z+1)} \left[1 + \frac{1}{(z+1)} + \frac{1}{(z+1)^2} + \dots \right] - \frac{3}{(z+1)}$$

$$- \frac{2}{3} \left[1 - \left(\frac{z+1}{3} \right) + \left(\frac{z+1}{3} \right)^2 + \dots \right]$$

$$\Rightarrow f(z) = \left[\frac{-1}{(z+1)} - \frac{1}{(z+1)^2} - \dots \right] - \frac{3}{(z+1)} - \frac{2}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z+1}{3} \right)^n$$

$$\Rightarrow f(z) = -\sum_{n=0}^{\infty} \left(\frac{1}{z+1}\right)^n - \frac{3}{(z+1)} - \frac{2}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z+1}{3}\right)^n.$$

(iii) $|z+1| > 13$

Q.4) Find the Laurent's Series and Taylor's Series which represent the function,

$$f(z) = \frac{z^2 - 1}{(z+2)(z+3)} \quad \text{when}$$

$$(i) |z| < 2 \quad (ii) 2 < |z| < 3 \quad (iii) |z| > 3$$

Solⁿ:

$$f(z) = \frac{z^2 - 1}{(z+2)(z+3)} = \frac{z^2 - 1}{z^2 + 5z + 6}$$

$$\begin{array}{r} z^2 + 5z + 6 \overline{) z^2 - 1} \quad 1 \\ \underline{z^2 + 6 + 5z} \\ (-) + 6 + 5z \\ \underline{-5z - 7} \end{array}$$

$$\Rightarrow f(z) = 1 - \frac{(5z+7)}{z^2+5z+6}$$

$$\Rightarrow f(z) = 1 - \frac{(5z+7)}{(z+2)(z+3)}$$

$$\frac{5z+7}{(z+2)(z+3)} = \frac{A}{(z+2)} + \frac{B}{(z+3)}$$

$$5z+7 = A(z+3) + B(z+2)$$

$$\text{for } z = -3, \\ b = 8$$

$$\text{for } z = -2, \\ A = -3$$

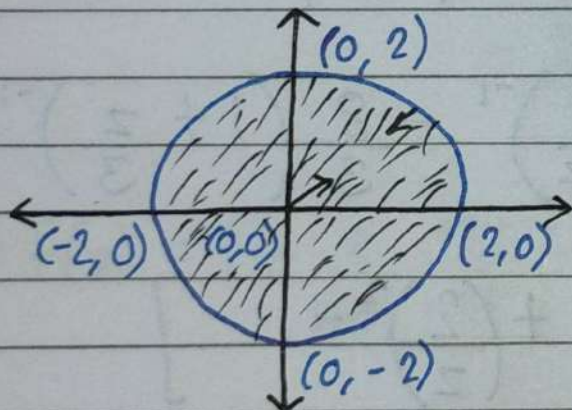
$$f(z) = 1 - \left[\frac{(-3)}{(z+2)} + \frac{8}{(z+3)} \right]$$

$$\Rightarrow f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

$$\Rightarrow f(z) = 1 + 3(z+2)^{-1} - 8(z+3)^{-1}$$

$$(i) \quad |z| < 2$$

$$\therefore |z| < 2$$



$$\left| \frac{z}{2} \right| < 1$$

$$\left| \frac{z}{3} \right| < \left| \frac{z}{2} \right| < 1$$

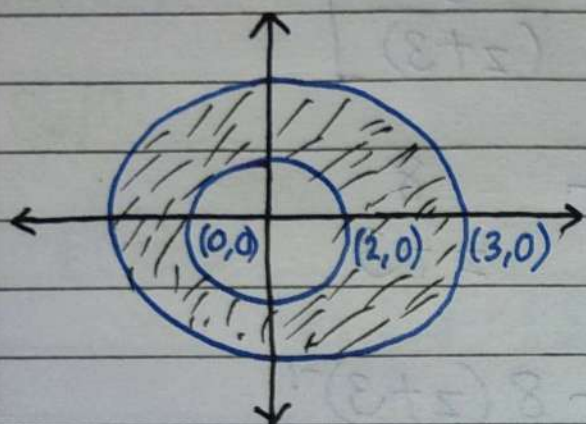
$$f(z) = 1 + \frac{3}{2} \left(1 + \frac{z}{2} \right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3} \right)^{-1}$$

$$\Rightarrow f(z) = 1 + \frac{3}{2} \left[1 - \frac{z}{2} + \left(\frac{z}{2} \right)^2 - \dots \right] \\ - \frac{8}{3} \left[1 - \frac{z}{3} + \left(\frac{z}{3} \right)^2 - \dots \right]$$

$$\Rightarrow f(z) = 1 + \frac{3}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$$

This is Taylor's series.

(ii) $2 < |z| < 3$



$$\because |z| > 2$$

$$\left| \frac{2}{z} \right| < 1$$

$$\because |z| < 3$$

$$\left| \frac{z}{3} \right| < 1$$

$$f(z) = 1 + \frac{3}{z} \left(1 + \frac{2}{z}\right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1}$$

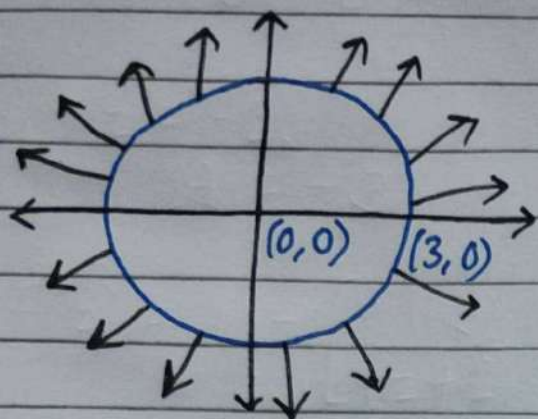
$$\Rightarrow f(z) = 1 + \frac{3}{z} \left[1 - \frac{2}{z} + \left(\frac{2}{z}\right)^2 - \dots \right]$$

$$- \frac{8}{3} \left[1 - \frac{z}{3} + \left(\frac{z}{3}\right)^2 - \dots \right]$$

$$\Rightarrow f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$$

This is ~~Taylor's~~ Laurent's series.

(iii) $|z| > 3 \Rightarrow 3 < |z| < \infty$



$$\therefore |z| > 3$$

$$\left| \frac{3}{z} \right| < 1$$

$$\left| \frac{2}{z} \right| < \left| \frac{3}{z} \right| < 1$$

$$f(z) = 1 + \frac{3}{z} \left(1 + \frac{2}{z} \right)^{-1} - \frac{8}{z} \left(1 + \frac{3}{z} \right)^{-1}$$

$$\Rightarrow f(z) = 1 + \frac{3}{z} \left[1 - \frac{2}{z} + \left(\frac{2}{z} \right)^2 \dots \right]$$

$$- \frac{8}{z} \left[1 - \frac{3}{z} + \left(\frac{3}{z} \right)^2 \dots \right]$$

$$\Rightarrow f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z} \right)^n - \frac{8}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3}{z} \right)^n$$

This is Laurent's series.